

TABLE 4—LINEAR-IN-MEANS MODELS OF PEER EFFECTS IN ACHIEVEMENT FOR HOUSTON
(OLS and 2SLS models using Grade-Level Evacuee Share as an instrument for peer achievement—elementary)

	OLS— Pre-Katrina only	OLS— all years	2SLS— first stage	2SLS— second stage	OLS model with peer achievement interacted with own quartile			
Dependent variable →	Avg. peer score	Avg. peer score	Evacuee share	Avg. peer score	Avg. peer score × quartile 1	Avg. peer score × quartile 2	Avg. peer score × quartile 3	Avg. peer score × quartile 4
	(1)	(2)	(3)	(4)	(5)			
Math	0.90*** (0.04)	0.79*** (0.04)	−1.61*** (0.50)	0.18 (0.22)	0.82*** (0.04)	0.77*** (0.05)	0.67*** (0.05)	0.58*** (0.07)
Observations	54,148	92,259	92,259	92,259	90,259			
F-test of joint equality					7.2***			
Reading	0.84*** (0.05)	0.70*** (0.05)	−1.13** (0.51)	0.11 (0.31)	0.84*** (0.06)	0.68*** (0.06)	0.59*** (0.08)	0.37*** (0.08)
Observations	43,053	73,918	73,918	73,918	73,918			
F-test of joint equality					10.5***			
	Reduced form with peer achievement interacted with own quartile				2SLS model with peer achievement interacted with own quartile—second stage instrumented with evacuee share × own quartile with lags			
Dependent variable →	Evacuee share × quartile 1	Evacuee share × quartile 2	Evacuee share × quartile 3	Evacuee share × quartile 4	Avg. peer score × quartile 1	Avg. peer score × quartile 2	Avg. peer score × quartile 3	Avg. peer score × quartile 4
	(6)				(7)			
Math	−0.75 (0.85)	−1.16** (0.51)	−0.99* (0.57)	0.73 (0.50)	0.43 (0.43)	0.70** (0.30)	0.62* (0.32)	−0.52 (0.46)
Observations	90,259				90,259			
F-test of joint equality	3.1**				1.5			
Reading	−0.42 (0.93)	0.12 (0.58)	0.15 (0.38)	0.18 (0.57)	0.30 (0.66)	−0.13 (0.70)	−0.15 (0.43)	−0.14 (0.48)
Observations	73,918				73,918			
F-test of joint equality	0.1				0.1			

Notes: Standard errors are provided in parentheses and clustered by school. Regressions cover 2003–2004 and 2006–2007 and include student’s race, gender, free/reduced price lunch status, grade-by-year fixed effects, school fixed-effects, and lagged achievement from the student’s most recent pre-Katrina lagged score interacted with indicators for the number of years between exams. In regressions 5, 6, and 7, all covariates are interacted with the student’s quartile to generate fully interacted models. TAKS scores are standard deviations of scale scores within grade and year using statewide means and standard deviations. When students have multiple scores for a single subject in a given year we use the lowest score. Quartiles are determined using the most recent pre-Katrina score available and are within grade and year across the district. Elementary is defined as any student in grades 3–5. Middle/High is any student in grades 6–11.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.